#1 IAM policy screenshots:

```
`
    "RoleName": "Lab03UploaderRole",
    "PolicyName": "Lab03UploaderPolicy",
    "PolicyDocument": {
        "Version": "2012-10-17",
        "Statement": [
            {
                "Effect": "Allow",
                "Action": [
                    "s3:PutObject"
                ],
                "Resource": "arn:aws:s3:::cloud-lab03-files-100995727963/shared.txt"
            }
        ]
    }
}
root@0392935df869:/workspace/lab03# aws iam get-role-policy \
  --role-name Lab03ViewerRole \
[ --policy-name Lab03ViewerPolicy \
  --no-cli-pager
{
    "RoleName": "Lab03ViewerRole",
    "PolicyName": "Lab03ViewerPolicy",
    "PolicyDocument": {
        "Version": "2012-10-17",
        "Statement": [
            {
                "Effect": "Allow",
                "Action": "s3:GetObject",
                "Resource": "arn:aws:s3:::cloud-lab03-files-100995727963/shared.txt"
            }
        ]
    }
}
root@0392935df869:/workspace/lab03#
```

Screenshot#2-App-Folders:

```
       <pre id="content"></pre>
[EOF.listen(PORT, () => console.log(`Viewer listening on ${PORT}`));;') +
 root@0392935df869:/workspace/lab03/viewer-app# node --check app.js
[ls -l
 total 8
 -rw-r--r-- 1 root root 1227 Oct  9 20:18 app.js
 -rw-r--r-- 1 root root  157 Oct  9 20:17 package.json
[root@0392935df869:/workspace/lab03/viewer-app# cd /workspace/lab03
 root@0392935df869:/workspace/lab03# ls -l uploader-app/
[ls -l viewer-app/
 total 8
 -rw-r--r-- 1 root root 2119 Oct  9 20:15 app.js
 -rw-r--r-- 1 root root  189 Oct  9 20:14 package.json
 total 8
 -rw-r--r-- 1 root root 1227 Oct  9 20:18 app.js
 -rw-r--r-- 1 root root  157 Oct  9 20:17 package.json
 root@0392935df869:/workspace/lab03#
```

Screenshot #3

```
root@0392935df869:/workspace/lab03# aws ec2 describe-launch-templates \
  --launch-template-names \
    Lab03-Uploader-Template \
    Lab03-Viewer-Template \
  --query "LaunchTemplates[*].[LaunchTemplateName,LaunchTemplateId,LatestVersionNumber]" \
  --output table \
  --no-cli-pager
-------------------------------------------------------------
|                DescribeLaunchTemplates                    |
+--------------------------+-----------------------+----+
|  Lab03-Uploader-Template |  lt-0becc4e32cd152081  |  1 |
|  Lab03-Viewer-Template   |  lt-0216805a7b479f9a2  |  1 |
+--------------------------+-----------------------+----+
root@0392935df869:/workspace/lab03#
```

Screenshot #4

```
  --output table \
  --no-cli-pager
-----------------------------------
| DescribeLaunchTemplateVersions  |
+---+-------------------------------+
|  4|  Uploader updated version     |
|  3|  Uploader updated version     |
|  2|  Uploader version 2           |
|  1|  Uploader version 1           |
+---+-------------------------------+
root@0392935df869:/workspace/lab03#
```

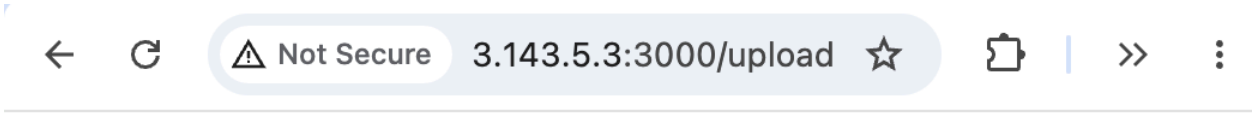
Screenshot #5

```
Both instances are running!

-------------------------------------------------------
|                  DescribeInstances                  |
+----------------------+----------+------------------+
|  i-09b55fa2c81847d7a |  running |  18.117.186.31   |
|  i-097746f2a4ced5062 |  running |  3.12.76.187     |
+----------------------+----------+------------------+

Lab 03 stack created successfully!
root@0392935df869:/workspace/lab03#
```

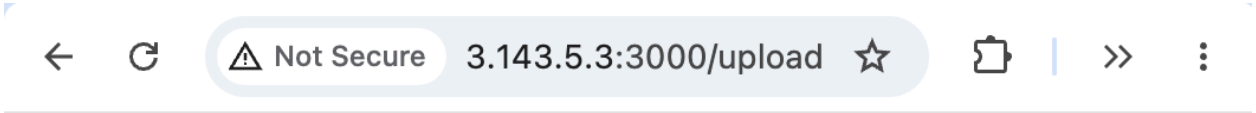
Screenshot #6 - upload is successful



Upload successful.

Upload another

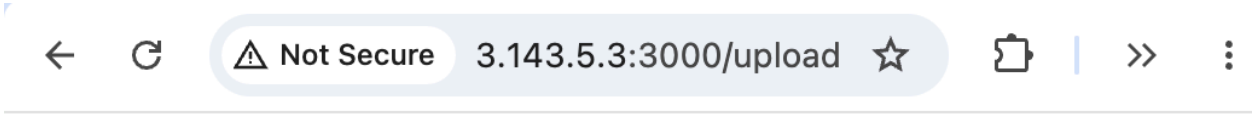
Screenshot#7 large file error message



Upload failed: file exceeds the 1MB limit.

Try again

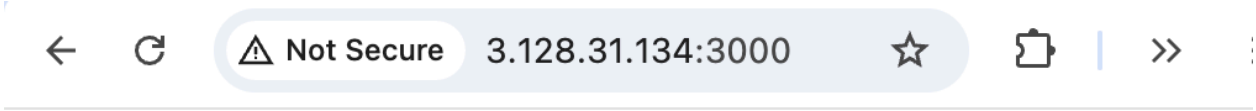
Screenshot#8 wrong file message



Upload failed: only .txt files are allowed.

Try again

Screenshot#9 Viewer page



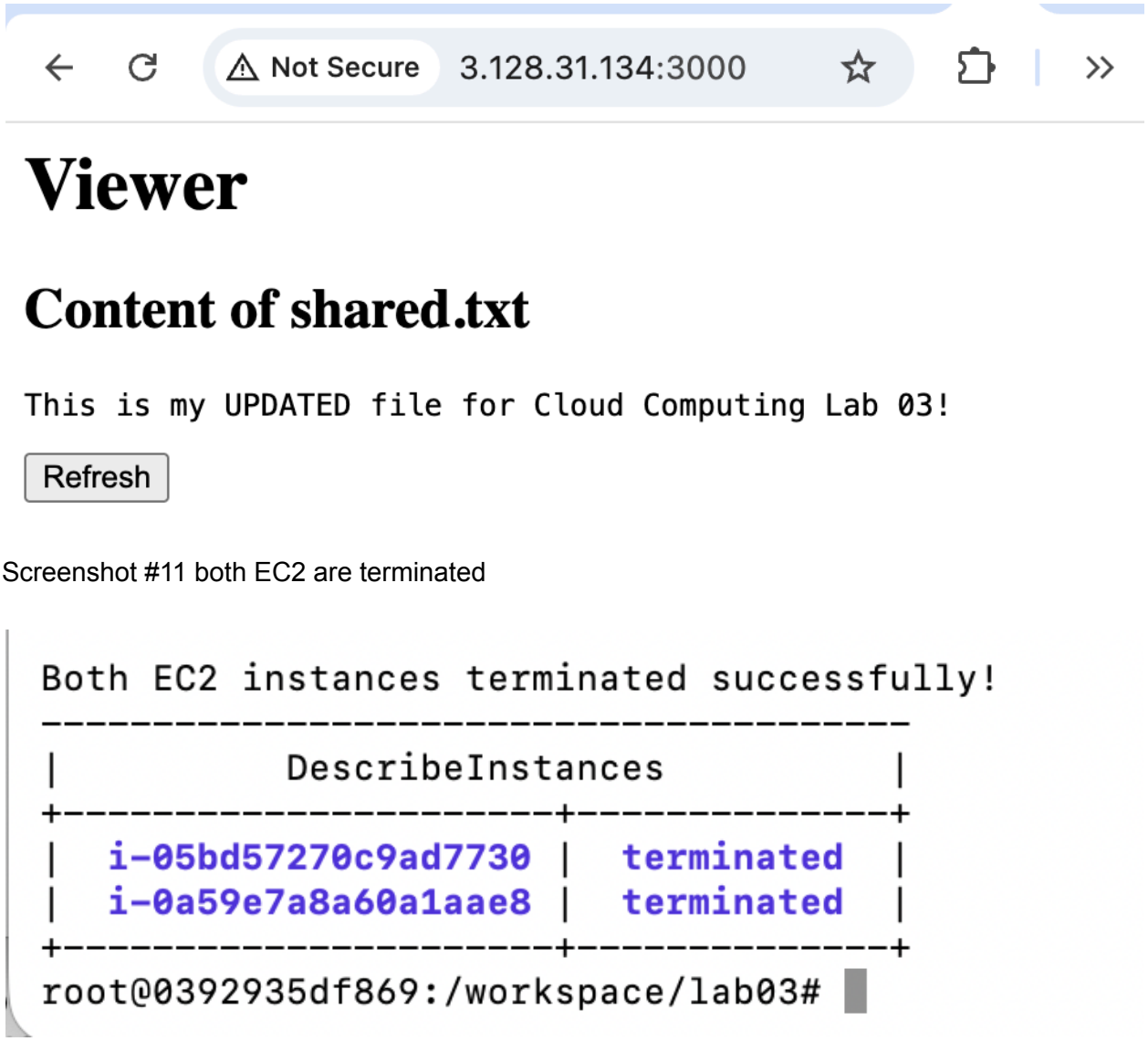
Viewer

Content of shared.txt

```
Hello! This is my first Cloud Computing Lab 03 upload.
```

Refresh

Screenshot #10 upload another txt file and refresh



Screenshot #11 both EC2 are terminated

```
Both EC2 instances terminated successfully!
-----------------------------------------
|          DescribeInstances            |
+----------------------+----------------+
|  i-05bd57270c9ad7730 |  terminated    |
|  i-0a59e7a8a60a1aae8 |  terminated    |
+----------------------+----------------+
root@0392935df869:/workspace/lab03#
```

Written part:
Why do the Uploader and Viewer use separate IAM roles?

The Uploader and Viewer use separate IAM roles to improve security. The Uploader only needs permission to upload the shared.txt file to S3, while the Viewer only needs permission to read it. Using separate roles makes sure each application has only the permissions it needs. This reduces security risks and prevents unnecessary access.

Why is the application code embedded in user-data?

Embedding the application code directly into user-data allows the EC2 instance to create the application files when it starts. This means the instance does not need Git or S3 credentials to download its own application code.